

David Sai

650-622-6855 | davidsai@ucla.edu | www.linkedin.com/in/david-sai/ | david-sai.github.io

EDUCATION

University of California, Los Angeles (UCLA)

Expected June 2027

Bachelor of Science in Computer Science

GPA: 3.94/4.0

Relevant Coursework: Deep Learning, Neural Networks, Data Structures and Algorithms, Software Construction, Operating Systems, Computer Architecture, Programming Languages, Object Oriented Programming

EXPERIENCE

Software Development Engineer Intern

June 2025 – Present

Amazon Web Services (AWS)

Seattle, WA

- Built and deployed a service using Amazon Bedrock, LangChain (RAG), and LLMs to automatically generate AWS IAM policies from client code, enhancing resource security and streamlining permissions management
- Automated the provisioning of service-linked roles (SLRs) for unsuspended and restored AWS accounts using Amazon SNS, SQS, and Lambda, saving internal service teams thousands of hours of manual integration work
- Collaborated with the Account Event Service team to transition the SLR provisioning workflow to an event-driven architecture, enabling near-instant policy updates upon account state changes

Technical Advisor Intern – Software Engineering

January 2025 – Present

Scale AI

Remote

- Trained generative AI models to tackle competitive programming and algorithmic problem-solving tasks
- Optimized LLM performance on real-world software engineering, focusing on open-source project contributions

Full Stack Software Engineer

September 2023 – Present

Daily Bruin (<https://bruinwalk.com/>)

Los Angeles, CA

- Developed new pages and database models for dorms and dorm reviews using Django, Javascript, Python, Docker and HTML/CSS on Bruinwalk, a review tool with 5.7 million website views from 50,000+ UCLA students annually
- Built new "saved pages" feature allowing thousands of students to plan classes more effectively

Information Systems Engineer Intern

June 2024 – September 2024

Genentech

Oceanside, CA

- Enhanced system security by developing, implementing, and deploying a Python script utilizing Selenium for web automation and BeautifulSoup for web scraping to verify and audit user system access.
- Reduced setup time of mini PCs for new HMIs by 60% by automating the configuration with a PowerShell script.

Software Engineer Intern

June 2023 – September 2023

Geothara

Remote

- Transcribed the text from images of documents using Tesseract and Python with 95% accuracy
- Preprocessed thousands of images for Optical Character Recognition using filters, transformations, and noise reduction methods in OpenCV, improving character recognition accuracy by over 15%

PROJECTS

Keystroke Prediction from sEMG Signals ([Research Paper Link](#))

January 2025 – March 2025

- Trained neural networks to predict keystrokes based on surface electromyography (sEMG) signals from the brain
- Explored different architectures including CNNs, RNNs, LSTMs, GRUs, Transformers, and hybrid architectures to optimize performance, achieving a character accuracy rate of over 75%
- Utilized PyTorch and NumPy to implement data augmentation, optimization, and other deep learning techniques

Local Find (Creative Labs) ([Github Link](#))

January 2024 – March 2024

- Designed and built Local Find, a web app for promoting local business and street vendors utilizing React, Typescript, Python, and Firebase, collaborated with a team of 12 UI/UX designers and software developers
- Led the integration of Google Maps API, enabling restaurant exploration and marker placement functionality.

TECHNICAL SKILLS

Languages and Frameworks: C++, Java, Python, SQL, HTML, CSS, Javascript, C#, Django, Typescript, React, MongoDB, Express, Node, Rust, Scheme, Lisp, Prolog, OCaml

Developer Tools: Git, VS Code, Visual Studio, IntelliJ, Eclipse, Unity, Android Studio, Docker, Cline

Design: Photoshop, Illustrator, Figma, Canva, UI/UX, Graphic Design, Premiere Pro